Logan Thomas

Mechanical Engineering · Seeking Summer 2027 Internship

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Education

University of California, Los Angeles · B.S. Mechanical Engineering · *GPA 4.00 / 4.00* Expected Jun 2029

Fall 2026: Project Lead, Bruin Space CubeSat · Researcher, Mechanics of Soft Materials Lab (liquid crystal elastomers)

Skills

CAD & Design: SolidWorks (CSWA), AutoCAD, GD&T / Tolerancing, DFM, Machinist Drawings, Mechanism Design
Analysis & Code: FEA (SolidWorks Simulation), Statics, MATLAB, Simulink / Simscape, Python (scikit-learn), C++
Fabrication: Waterjet, Laser Cutting, FDM 3D Printing, Silicone Molding, Sheet Metal Forming, Soldering
Test & Electronics: Instrumented Testing, Design of Experiments, Root Cause Analysis, Harness Build, Arduino / IMU

Experience

DOER Marine · Engineer Intern · *Manned Research Submersible* Jun 2026 – Sep 2026

- **Designed a moonpool transducer deployment rig** from rough concept to manufacturable assembly, specifying the lead screw, bearings, and guide shafts, and isolating aluminum from stainless hardware against galvanic corrosion in seawater.
- **Analyzed the rig's load-bearing frame** with hand calculations and self-taught FEA after the director of engineering flagged it as thin, comparing three beam setups and showing that closing an unsupported span cut peak stress 74% on stock C-channel.
- **Built an instrumented thermal test rig** to benchmark two battery-wall cooling designs, showing the solid-wall design outperformed, and wrote the formal test procedure and calibrated a Simscape model against the data.
- **Built and debugged subsea ethernet and CAN harnesses** for a pressure test, tracing a dead link to a manufacturer schematic that contradicted the connector's physical wire order, then logged both links through pressure testing to 2,000 PSI.
- **Produced the battery enclosure drawing package**, toleranced for machining, and caught an interference in a vendor part.
- **Relocated the accumulator tank**, designed its bracket, and routed AC plumbing through the spherical hull.

I²BL, UCLA Bioelectronics Lab · Engineering Research Intern · *Ferrobotic Viral Testing Platform* Jan 2026 – Jun 2026

- **Developed a mass-producible molding process** for self-sealing silicone ports, root-causing air-entrapment defects.
- **Engineered a modular optical enclosure** with a sliding cartridge held to sub-millimeter vertical tolerance, keeping the ferrobot's magnet close enough to move ferrofluid reliably.
- **Designed and printed a working cross-spherical gear mechanism** to explore two-axis tilt control.
- **Wrote C++ calibration code for an MPU-6050** to compute real-time tilt and correct sensor drift in flight.

Independent research project: machine-learning model of microfluidic dispenser behavior

- **Built a machine-learning pipeline end to end**, from experimental design and laser-cut chip fabrication through 369 trials and MATLAB image analysis, mapping dispenser geometry to droplet output.
- **Trained a random-forest model** to predict droplet volume, diagnosed overfitting through cross-validation, and turned it into an inverse-design tool for choosing dispenser geometry.

Projects

Self-Deploying Laptop Stand · Personal Project · *Mechanism Design* Jun 2026 – Sep 2026

- **Derived the linkage statics symbolically** before opening CAD, proving no equilibrium exists at any deployed angle, so the hard stop, not the spring, has to carry the load.
- **Moved the load path to a compression flap**, cutting FEA deflection from 2 mm to 0.1 mm, then cut mass 46% with ribs.
- **Loaded the stand to failure at 10× design load**, tracing it to a friction joint slipping, not a material failure.
- **Ran a 12-sample PETG bend test** across infill patterns and densities, found pattern drives stiffness about 3× more than density, and thickened walls instead of raising infill.

X1 Robotics · Internal Arm Engineer · *Autonomous Spherical Robot, Student Team* Sep 2025 – Jun 2026

- **Modeled and assembled the yaw motor housing** in SolidWorks and refined it for 3D-printing manufacturability.
- **Prototyped a rod-and-sleeve coupling** to decouple head rotation from the drivetrain, finding its sliding fit impractical to machine as one part and informing a redesign to a separate keyed feature.

Electric Guitar Builds · Personal Projects · *Woodworking & Analog Electronics* Jun 2024 – Jul 2025

- **Machined a custom guitar body from raw stock** with template routing, laying out cavities around the full control circuit.
- **Designed the control circuit from scratch**, with a kill switch, coil split, and volume and tone controls, star-grounded for low noise, and set up a Floyd Rose tremolo against string tension.